

Finetuning a Character-level Cantonese Text-to-Speech System

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1 Introduction

This study develops a Cantonese Text-to-Speech (TTS) system targeting the Liwan (Canton) dialect, utilizing the Bert-VITS2 framework, an enhanced iteration of the VITS2 architecture. The training dataset is derived from the Liwan sub-corpus of the China's National Project of Protection of Language Resources (CPLR)¹. The model is trained using WAV audio files and TextGrid annotations, which are aligned via HKCantonese Models and the Montreal Forced Aligner (MFA). The synthesized speech will be evaluated through Mean Opinion Score (MOS) and Comparative Mean Opinion Score (CMOS) by native Cantonese speakers, focusing on 3 dimensions: *pronunciation, articulation, comprehension, naturalness, and voice quality*.

Cantonese is a Sinitic language widely spoken in Chinese communities across Mainland China, Hong Kong, Malaysia, and the United States. It is also considered a low-resource language because it lacks official status in any region other than Hong Kong. However, Snow (2010) characterizes Hong Kong as a case of modern diglossia, where English and Standard Written Chinese function as 'high' varieties over Cantonese. This functional hierarchy accounts for the scarcity of Cantonese language resources.

2 Materials and Modeling Methodology

2.1 HKCantonese_models (Cantonese pretrained model) by *chenchenzi*² and PyCantonese

The HKCantonese Model is an ASR (Automatic Speech Recognition) system developed by Xu (2023). It comprises pre-trained acoustic models for Hong Kong Cantonese alongside a Chinese character-to-IPA mapping lexicon dictionary *cv19_val_lexicon_v3.dict*. This lexicon includes detailed parameters such as confidence scores, durations, feature weights, and pause predictions. Notably, the model also incorporates the phonetic adaptation of English loanwords. Utilizing the Montreal Forced Aligner (MFA), we integrated the pre-trained acoustic model and the lexicon dictionary to generate TextGrid files. This pipeline enables the automated alignment of phonetic labels with raw WAV audio based solely on character-level transcriptions.

PyCantonese (Lee, 2015) was utilized for automatic Jyutping transcription, a critical step in generating precise TextGrid files. This ensured high-quality forced alignment essential for effective model training.

2.2 Bert-VITS2 Framework Training and HiFiGAN Vocoder Synthesization

In this study, we employed Bert-VITS2 for the Text-to-Speech (TTS) training pipeline. The preprocessing phase involved converting raw WAV audio and TextGrid annotations into mel-spectrogram parameters to facilitate downstream training. The model was trained for 9 epochs with a learning rate of $1 * e^{-3}$, reaching a stable final loss of approximately 33. The resulting model was saved as G900.pth, accompanied by its corresponding config.json file, which contains the necessary hyperparameters and architectural specifications.

During the inference (synthesis) stage, the model requires eight input parameters: phoneme sequences (Chinese characters/Jyutping), input sequence lengths, speaker IDs, tone labels, language categories, and BERT-based semantic embeddings. The system first generates the mel-spectrogram, which is subsequently transformed into high-fidelity speech using the HiFi-GAN vocoder.

3 Assessment Methodology and Result

3.1 Mean Opinion Score (MOS) and Comparative Mean Opinion Score (CMOS)

¹ State Language Commission of People's Republic of China. (2015). *China's National Project for the Protection of Language Resources*. Accessed 31/03/2026. (Available at <https://zhongguoyuyan.cn/>)

² Xu, Chenzi. (2023). *HKCantonese Model*. Accessed 31/03/2026. (Available at Github: https://github.com/chenchenzi/HKCantonese_models/tree/main)

The evaluation of synthesized audio is multi-dimensional. Viswanathan et al. (2005) stated that the quality of Text-to-Speech (TTS) systems can only be effectively assessed through reliable and valid listening tests. Currently, the Mean Opinion Score (MOS) remains the standard measure for evaluating synthesized speech quality (ITU, 1994).

In this study, the test set consists of prompts including 30 isolated Chinese characters (Hanzi) and 3 full colloquial sentences selected from the China’s National Project for the Protection of Language Resources (CPLR). Two native Cantonese speakers served as evaluators, providing scores on a 5-point scale (with a 0.5-point increment) for the 30 isolated characters. Subsequently, the synthesized audio and ground-truth recordings (from CPLR) of the 3 sentences were paired and randomized for a Comparative Opinion Score evaluation, ranging from -2 to +2 (with a 1-point increment). The final results are presented as average scores, referred to as MOS and CMOS in the following sections. Evaluators are also required to assign problem tags—including *pronunciation*, *articulation*, *comprehension*, *naturalness*, and *voice quality*—to identify specific issues when a low score is given. The model is expected to achieve a MOS of over 3.5 and a CMOS within the range of (-0.5, 0) to meet our performance expectation.

Sequence	System A (randomized)	System B (randomized)	Comparative Score	Tags	Comments
1	[Audio]	[Audio]			
2	[Audio]	[Audio]			
3	[Audio]	[Audio]			

Figure 1. A CMOS evaluation online table sample

3.2 Results

Although evaluation tasks and the CPLR dataset were prepared, the synthesis phase remains incomplete. Bert-VITS2’s heavy optimization for Mandarin makes adapting it for pure Cantonese—rather than a hybrid output—require extensive architectural adjustments beyond this stage.

4 Discussion and Limitations

A key improvement is the use of pycantonese for Jyutping-based G2P conversion, which enables accurate modeling of tones and syllable structures. Compared to earlier character-level approaches, this significantly enhances linguistic consistency in the input representation. However, the system still relies on Mandarin-based BERT embeddings for Cantonese, which creates a mismatch between semantic and phonological representations. While this workaround allows the model to function, it may limit naturalness and prosodic accuracy. In addition, the use of MFA introduces dependency on alignment quality that is sensitive to text normalization and segmentation choices.

Several limitations affect the robustness of the system. First, the reuse of Mandarin BERT for Cantonese limits linguistic precision and may impact synthesis quality. Second, the preprocessing pipeline (alignment, slicing, and filtering) is complex and prone to error propagation, requiring manual intervention. Third, phoneme segmentation is manually defined and may introduce inconsistencies in edge cases.

While the current synthesis phase faced technical impediments, this study has yielded a profound understanding of the intricacies involved in TTS fine-tuning for tonal languages. This exploration has clarified the limitations of BERT-based cross-lingual adaptation and the critical role of phonetic alignment. Moving forward, we aim to implement more advanced, lightweight architectures such as Matcha-TTS to optimize low-resource language modeling. Our future work will extend beyond the training phase to encompass full speech synthesis and systematic evaluation.

5 References

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6 Code File

Project repository uploaded to Github: https://github.com/hanlincl/Finetuning_bertvits2_for_cantonese_tts

7 CRediT Author Statement

Zhou, Wanqiong: Data curation, Methodology, Formal Analysis, Validation, Evaluation

Li, Haoyu: Investigation, Writing – Review & Editing, Project Administration.

Chen, Hanlin: Writing – Original Draft Preparation, Resources, Conceptualization, Evaluation